

Functional CLT by the δ -method — general skeleton

Shaded **n** = slot that changes with the problem. Page 2 fills every slot for four different routes. **Classify before you differentiate:** three of the nine map shapes below need no domain at all.

FOUR DECISIONS — keep them apart. **index set** (where t lives) **FORCED** by the weak-convergence input $\cdot$ **value range** (a, b) (where $\theta(t)$ lives) **FREE** — and never appears in your answer $\cdot$ **tangential set** D_0 **FORCED** by the lemma + the paths of the limit $\cdot$ **domain type** (value cut / index cut / BV bound / none) **FORCED** by which lemma.

STEP 0 — CLASSIFY THE MAP. Find your shape; the rest of the row is the answer. Compose with Thm 1.16.

map shape	cite	domain	D_0	derivative
bounded linear : $\int Fw$, $\int_0^t F$, $F \mapsto F(t_0)$, restriction, $F \mapsto (F, F)$, $H \mapsto H(\cdot, \infty)$	Def. 1.12 direct; Prop. 1.2(1),(2)	whole space — <i>none</i>	everything	$\phi'_\theta = \phi$ (two lines)
bounded bilinear : $(f, g) \mapsto fg$	one line, $\ fg\ \leq \ f\ \ g\ $	whole space — <i>none</i>	everything	$h_1g + fh_2$; cross term $O(t^2)$
pointwise $\phi(F(\cdot))$	Lem. 1.10(2)	value cut (a, b) missing every singularity of ϕ'	everything	$\phi'(F(\cdot))h(\cdot)$
inverse $F^{\leftarrow}$	Lem. 1.10(1)	index cut $[u_1, u_2] \subset (0, 1)$; $f > 0$ on $[a, b]$	$C[a, b]$	$-(h/f) \circ F^{-1}$
ratio of integrals $\int \psi(F_1) dF_2$	Prop. 1.2(3) Lem. 1.11	denominator $\neq 0$ (open) $BV_M \times BV_M$, and $\psi \in C^2$ on the range of F_1	everything BV directions in 2nd slot	quotient rule on L_1, L_2 3 terms: boundary + by-parts + chain
product integral $\prod(1 + dF)$	Lem. 1.12 (1.13 matrix)	$BV_M(0, b]$ — not open	BV directions	$\int_{(0, \cdot]} \phi(F)(0, u-) \phi(F)(u, \cdot] dh$
composition $F_2 \circ F_1$	Lem. 1.10(3)	F_2 uniformly Fréchet diff. over range of F_1	$\ell^\infty \times UC$	$h_2 \circ F_1 + F'_{2, F_1}(h_1)$
finite-dimensional $\mathbb{R}^k \rightarrow \mathbb{R}^m$	ordinary total differentiability	open set avoiding the kink / zero	$\mathbb{R}^k$	the Jacobian

STEP 1 — DECOMPOSE. $T = \mathbf{1}$, a composition of **2** catalogue maps. Chain rule, Thm 1.16. *Never differentiate by hand what the catalogue gives.*

STEP 2 — FIX THE DOMAIN, per stage, in this order. bad set of $\phi' = \mathbf{3}$; range of θ on the index set = **4** (use monotonicity); cut with slack $\delta = \mathbf{5}$, giving $D_\phi = \mathbf{6}$, of type **7**. *Slack, because $\theta + t_n h_n$ must lie in D_ϕ too: write strict inequalities so D_ϕ is open and this is automatic. If it cannot be open (BV), say so explicitly.*

STEP 3 — CITE AND ASSEMBLE. Per stage cite **8**; the tangential set is **9**. For a value cut, verify the two hypotheses at once: $\sup |\phi'| < \infty$ and $\sup |\phi''| < \infty$ on (a, b) , the second giving Lipschitz $\Rightarrow$ uniformly continuous ϕ' . Chain rule gives $\phi'_\theta(h) = \mathbf{10}$. *Proof-read: linear in h , with no h^2 , no $|h|$, no h at an h -dependent point.* Tangency bookkeeping: stage 2 must be differentiable tangentially to $\phi'_1(D_0)$, not to D_0 .

STEP 4 — ESTIMATOR LIES IN D_ϕ . $\mathbb{P}(\mathbf{11}) \rightarrow 1$. Monotone estimator $\Rightarrow$ the sup is an endpoint evaluation $\Rightarrow$ Glivenko–Cantelli or the LLN. For a BV domain, check *total variation*, not the supremum.

STEP 5 — WEAK-CONVERGENCE INPUT. By **12**: $\sqrt{n}(\mathbf{13} - \mathbf{14}) \rightsquigarrow \mathbf{15}$ in $\ell^\infty(\mathbf{16})$, tight, with paths in D_0 because **17**. *Joint needed?* **18** — if two processes enter the map, marginal convergence is *not* enough: union the Donsker classes, or use independence of two samples, or Slutsky if one limit is constant.

STEP 6 — δ -METHOD, Thm 1.17, $r_n = \sqrt{n}$. $\sqrt{n}(T(X_n) - T(\theta)) \rightsquigarrow \phi'_\theta(\mathbf{15}) = \mathbf{19}$. The generic direction h of Step 3 is instantiated at the limit process; linearity of ϕ'_θ is what keeps it Gaussian.

STEP 7 — COVARIANCE AND CHECK. $c(s, t) = \mathbf{20}$; for a *single-stage pointwise* transform only, this is $\phi'(F(s))\phi'(F(t))[F(s \wedge t) - F(s)F(t)]$ using $\text{Cov}(B(u), B(v)) = u \wedge v - uv$ and $F(s) \wedge F(t) = F(s \wedge t)$. Check **21**: set $s = t$ or specialise a parameter and recover a result you already know.

What loses marks. Cutting the index set when the *values* were the problem $\cdot$ marginal instead of joint convergence $\cdot$ never checking $X_n \in D_\phi$ $\cdot$ a derivative that is not linear in h $\cdot$ claiming tangency to all of D when the lemma gives C or UC $\cdot$ δ or a, b surviving into the final covariance $\cdot$ treating BV_M as open $\cdot$ a fifteen-minute ε - δ argument for something Lemma 1.10(2) gives in one line.

Choosing δ

The only genuinely free choice — and the one that must disappear again.

The three constraints. The right endpoint b must be (i) strictly below the singularity, or φ' is unbounded and the hypothesis fails; (ii) strictly above $\sup_t \theta(t)$, since you differentiate *at* θ , so θ must lie in the domain; (iii) far enough above it that $\theta + t_n h_n$ and X_n also fit, and $\|t_n h_n\|_\infty \rightarrow 0$, so any fixed gap will do. **Hence: any number strictly between $\sup \theta$ and the singularity works, and they all give the same derivative and the same limit law.** Halving is merely an explicit way to name one. **The picture, with numbers.** Say $F(\tau) = 0.8$. Gap to the singularity = 0.2, so $\delta = 0.1$ and $b = 1 - \delta = 0.9$:

$$0 \underbrace{\hspace{1.5cm}}_{\text{range of } F} = 0.8 \underbrace{\hspace{1.5cm}}_{\delta: \text{ slack for } F_{(n)}, t_n h_n} = 0.9 \underbrace{\hspace{1.5cm}}_{\delta: \text{ safety from } \varphi'} = 1 \varphi' = \infty$$

Writing $b = 1 - \delta$ rather than 0.9 does two jobs in one symbol: it is visibly below 1 (so $|\varphi'| \leq 1/\delta$, $|\varphi''| \leq 1/\delta^2$) and visibly above $F(\tau) = 1 - 2\delta$ (so there is room). The *other* endpoint, -1 , is arbitrary — φ' has no trouble on the left, so $(-\infty, 1 - \delta)$ works just as well.

φ	bad set	cut	δ
$-\log(1-x)$ hazard	$\{1\}$	$(a, 1 - \delta)$, $a < 0$ free	$\frac{1}{2}(1 - F(\tau))$
$1/x$ reciprocal survival	$\{0\}$	(δ, ∞)	$\frac{1}{2}S(\tau)$, S non-increasing
$\sqrt{x}$	$\{0\}$	(δ, b) , b free	$\frac{1}{2}F(t_0)$, needs $F(t_0) > 0$
$\log x$ log-quantile	$\{0\}$	(δ, ∞)	$\frac{1}{2}q_\alpha$ — on the <i>data</i> scale, not $[0, 1]$
$\log \frac{x}{1-x}$ logit	$\{0, 1\}$	$(\delta, 1 - \delta)$	$\frac{1}{2} \min\{F(t_0), 1 - F(\tau)\}$ — the <i>smaller</i> gap
x^2	none	(a, b) , both finite	no singularity, but $\varphi' = 2x$ still needs a finite b
$ z $	$\{0\}$, a <i>kink</i>	$\{ z > \delta\}$	$\frac{1}{2} \varphi(t_0) $; needs $\varphi(t_0) \neq 0$ <i>assumed</i>
inverse, linear, BV maps	—	no value cut at all	no δ — don't invent one

Tips. Read $\sup \theta$ and $\inf \theta$ off the *endpoints* — F, S, Λ and q are all monotone, so the sup is an evaluation, never a real supremum. Cut only the end where trouble sits; cutting a safe end is a visible over-correction. Bound φ' and φ'' in terms of δ in the same line — that is both hypotheses of Lemma 1.10(2) at once. If the bad point lies *inside* the range of θ , no δ exists and no cut will save you: the statistic is genuinely non-differentiable there, and the question must be assuming it away (as $\varphi(t_0) \neq 0$ does for $|\cdot|$) — say so explicitly. Fix δ in one line, use it three times, and never mention it again: **if δ or a, b appear in your final covariance, you have localised the wrong object.**

Your turn. Pick two maps from your own notes, and for each write the four things: bad set of φ' , range of θ (use monotonicity), the cut, and δ . If a map has no singularity, write “no cut” and move on — that is a correct answer, not a gap.

The same seven steps, four different routes

Cover the columns and fill them in. Note how little of the work is the δ -method itself: rows **3–9** are where the routes actually differ.

	Cumulative hazard (Ex. 15)	Mean excess $e = A \cdot (1/S)$	L -statistic $\int w q_u du$	Kaplan–Meier $\hat{S} = \prod(1 - d\hat{\Lambda})$
1 2 stages	one: $\Lambda = \phi(F)$ pointwise	two: reciprocal of S , then product	two: inverse, then a linear functional	one, after precomposing with the linear $\Lambda \mapsto -\Lambda$
8 cite	Lem. 1.10(2)	Lem. 1.10(2) + bilinear	Lem. 1.10(1)(i) + Prop. 1.2(1)	Lem. 1.12
3 bad set of ϕ'	$\{1\}$	$\{0\}$, and only for the denominator	none — the map is not a value transform	none — the issue is oscillation, not values
4 range of θ	$[F(0), F(\tau)]$, $F(\tau) < 1$	S non-increasing, so $\geq S(\tau) > 0$	on the <i>data</i> scale, not $[0, 1]$	—
5 6 D_ϕ	$\delta = \frac{1}{2}(1 - F(\tau))$; $\{h : h([0, \tau]) \subseteq (-1, 1 - \delta)\}$	$\delta = \frac{1}{2}S(\tau)$; $\ell^\infty \times \{h : h > \delta\}$ — numerator unconstrained	restrictions of dfs to $[a, b] = [F^{-1}(u_1) - \varepsilon, F^{-1}(u_2) + \varepsilon]$	$BV_M(0, b]$ with $M = 2\Lambda(b)$
7 type	value cut, open	value cut in one slot, open	index cut: $[u_1, u_2] = \text{supp } w \subset (0, 1)$	total-variation bound — not open
9 D_0	everything; no tangency needed	$\ell^\infty \times \ell^\infty$	$C[a, b]$, non-negotiable	directions of bounded variation — state this
10 $\phi'_\theta(h)$	$h/(1 - F)$	$(h_1 - e h_2)/S$	$-\int w(u) h(q_u)/f(q_u) du$	$-S(t)h(t)$ for continuous Λ
11 in D_ϕ	$F_{(n)}$ monotone $\Rightarrow \sup = F_{(n)}(\tau) \rightarrow F(\tau)$ (G–C)	$S_{(n)}$ monotone $\Rightarrow \inf = S_{(n)}(\tau) \rightarrow S(\tau)$	automatic: $F_{(n)}$ is always a df	<i>total variation</i> $= \hat{\Lambda}(b) \rightarrow \Lambda(b) < M$
12 input	Thm 1.22	Thm 1.19 (bracketing/Donsker)	Thm 1.22 on $[a, b]$, or Thm 1.23 for $\hat{q}$	whatever gives $\sqrt{n}(\hat{\Lambda} - \Lambda) \rightsquigarrow \mathbb{G}$
18 joint?	no	yes — union $\{(z - u)^+\}$ with $\{1(z > u)\}$; both Donsker, finite union is Donsker	no	no
19 limit	$B \circ F/(1 - F)$	$(\mathbb{G}_1 - e \mathbb{G}_2)/S$	$-\int w(u)B(u)/f(q_u) du$	$-S \cdot \mathbb{G}$
20 cov.	$\frac{F(s \wedge t) - F(s)F(t)}{(1 - F(s))(1 - F(t))}$	from $\text{Cov}(\mathbb{G}_i(u), \mathbb{G}_j(v)) = E[g_{i,u}g_{j,v}] - E g_{i,u} E g_{j,v}$	$\iint \frac{w(u)w(v)(u \wedge v - uv)}{f(q_u)f(q_v)} du dv$	$S(s)S(t) \text{Cov}(\mathbb{G}(s), \mathbb{G}(t))$
21 check	$s = t$ gives $F/(1 - F)$: Thm 1.12	marginal convergence would not have sufficed — say so	$w \equiv 1$ gives the sample mean and $\text{Var}(Z)$	sign: the lemma is for $\prod(1 + dF)$, so $S = \phi(-\Lambda)$

Three more, in one line each. **Expected shortfall** $(1 - \alpha - \delta)^{-1} \int_\alpha^{1-\delta} q_u du$: linear and bounded, so rows **3–7** are empty, $\phi'_q = \phi$, and all the work is Thm 1.23. If you find yourself choosing (a, b) here you have misread the map. **Empirical copula / ROC**: marginalise (linear) $\rightarrow$ invert (Lem. 1.10(1)) $\rightarrow$ compose (Lem. 1.10(3)); the new condition is tangency — the direction perturbing the *outer* function must be uniformly continuous, which is what continuity of $\partial H/\partial x, \partial H/\partial y$ buys. Two samples $\Rightarrow$ independence gives the joint limit for free. $\int \psi(F(s-)) dF(s)$: precompose with the diagonal $F \mapsto (F, F)$ — linear, hence free — then Lem. 1.11 and add both partials. Three terms, or you differentiated only one slot. $M = 1$ works, since a df has total variation 1.

WHAT THE HAZARD COLUMN LOOKS LIKE WRITTEN OUT. Seven steps, nine sentences. Nothing else is expected of you.

Theorem. Let (Z_n) be iid with continuous df F and fix τ with $F(\tau) < 1$. Then $\sqrt{n}(\Lambda_{(n)} - \Lambda) \rightsquigarrow (B \circ F)/(1 - F)$ in $\ell^\infty([0, \tau])$, a centred Gaussian process with covariance $c(s, t) = [F(s \wedge t) - F(s)F(t)]/[(1 - F(s))(1 - F(t))]$.

Proof. [1] Since $\Lambda = -\log(1 - F)$, we have $\Lambda = \phi(F)$ and $\Lambda_{(n)} = \phi(F_{(n)})$ for the pointwise map $\phi(h) = \varphi \circ h$, $\varphi(x) = -\log(1 - x)$: a single stage, so no chain rule is needed. [2] φ' is singular only at 1, and F is non-decreasing with $F(\tau) < 1$, so put $\delta = \frac{1}{2}(1 - F(\tau))$ and $D_\phi = \{h \in \ell^\infty([0, \tau]) : h([0, \tau]) \subseteq (-1, 1 - \delta)\}$, which is open in $\|\cdot\|_\infty$. [3] On $(-1, 1 - \delta)$ we have $|\varphi'(x)| = 1/(1 - x) \leq 1/\delta$ and $|\varphi''(x)| = 1/(1 - x)^2 \leq 1/\delta^2$, so φ' is bounded and, being Lipschitz by the mean value theorem, uniformly continuous. Lemma 1.10(2) therefore applies: ϕ is Hadamard differentiable at F tangentially to all of $\ell^\infty([0, \tau])$, with $\phi'_F(h) = h/(1 - F)$, which is linear and satisfies $\|\phi'_F(h)\|_\infty \leq \delta^{-1}\|h\|_\infty$. [4] $F_{(n)}$ is non-decreasing, so $\sup_{[0, \tau]} F_{(n)} = F_{(n)}(\tau) \rightarrow F(\tau) = 1 - 2\delta$ a.s. by Glivenko–Cantelli; hence $\mathbb{P}(F_{(n)} \in D_\phi) \rightarrow 1$ and $\Lambda_{(n)}$ is well defined on $[0, \tau]$ with probability tending to one. [5] By Theorem 1.22, $\sqrt{n}(F_{(n)} - F) \rightsquigarrow B \circ F$ in $\ell^\infty([0, \tau])$, tight; only one process enters the map, so no joint convergence is required. [6] Theorem 1.17 with $r_n = \sqrt{n}$ gives $\sqrt{n}(\Lambda_{(n)} - \Lambda) \rightsquigarrow \phi'_F(B \circ F) = (B \circ F)/(1 - F)$, Gaussian because ϕ'_F is linear. [7] Finally $\text{Cov}(B(u), B(v)) = u \wedge v - uv$ and $F(s) \wedge F(t) = F(s \wedge t)$ by monotonicity give $c(s, t)$ as stated. $\square$

Check: at $s = t$ this is $F(t)/(1 - F(t))$, the pointwise variance of Theorem 1.12. Note that δ has disappeared, as it must.

Order of operations, memorised: classify $\rightarrow$ bad set $\rightarrow$ range $\rightarrow$ cut with slack $\rightarrow$ cite $\rightarrow$ chain rule $\rightarrow$ estimator in domain $\rightarrow$ weak input (joint?) $\rightarrow$ Thm 1.17 $\rightarrow$ covariance $\rightarrow$ sanity check. **The slack δ and the endpoints a, b never appear after Step 3.**